

NAME : SHAH FAISAL

ROLL NO : 23I-0058

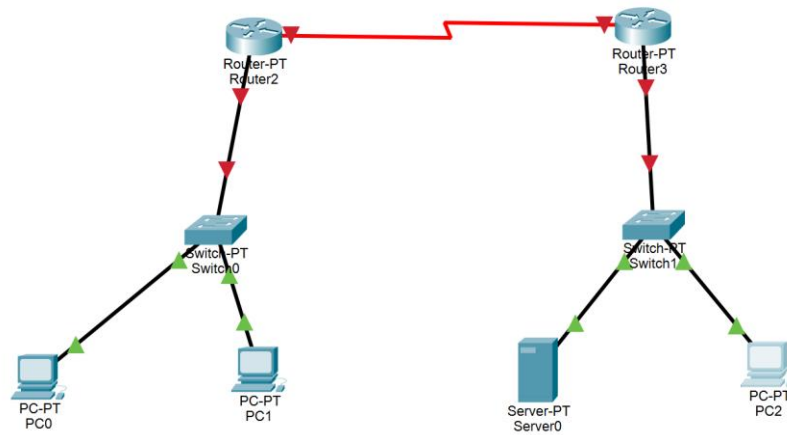
SECTION : BSE – 6A

Computer Network Lab 10

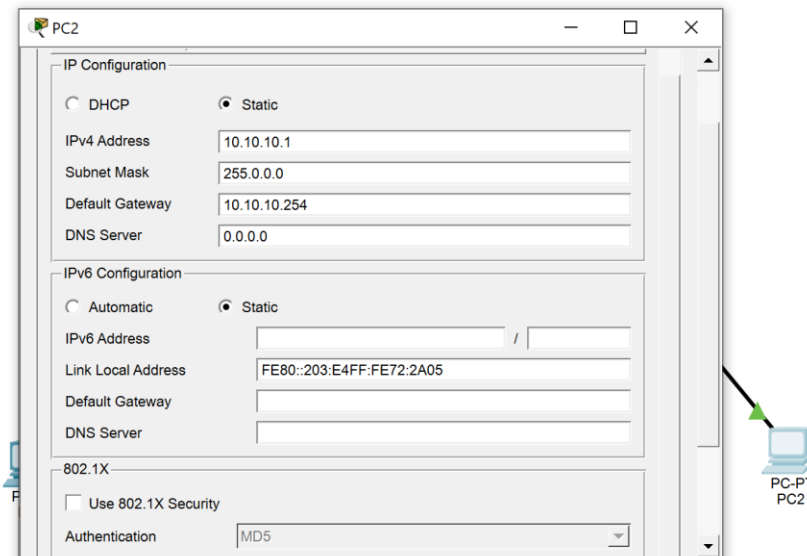
BSSE SPRING 2026

LAB 10

Task 1: Static NAT (10 marks)



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The screenshot shows the configuration window for PC2. The IP Configuration section is set to Static, with the following values:

- IPv4 Address: 10.10.10.1
- Subnet Mask: 255.0.0.0
- Default Gateway: 10.10.10.254
- DNS Server: 0.0.0.0

The IPv6 Configuration section is also set to Static, with the following values:

- IPv6 Address: (empty)
- Link Local Address: FE80::203:E4FF:FE72:2A05
- Default Gateway: (empty)
- DNS Server: (empty)

The 802.1X section shows the Use 802.1X Security checkbox is unchecked, and the Authentication dropdown is set to MD5.

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C-PT
PC0

Physical
Config
Services
Desktop
Programming
Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP
- PRP

File Name:

```

<html>
<center><font size='+2' color='blue'>Cisco Packet Tracer</font></center>
<hr>Welcome to Cisco Packet Tracer. Opening doors to new
opportunities. Mind Wide Open.
<p>Quick Links:
<br><a href='helloworld.html'>A small page</a>
<h1>Shah Faisal, 23I-0058, BSE-6A, Lab 10</h1>
<br><a href='copyrights.html'>Copyrights</a>
<br><a href='image.html'>Image page</a>
<br><a href='cscoptlogo177x111.jpg'>Image</a>
</html>
          
```

E NETWORK
NETWORK

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PC-PT
PC0

al
Config
CLI
Attributes

GLOBAL

Settings

rhythm Settings

ROUTING

Static

RIP

INTERFACE

Ethernet0/0

Ethernet1/0

Serial2/0

Serial3/0

Ethernet4/0

Ethernet5/0

Serial2/0

Port Status ☒ On

Duplex Full Duplex

Clock Rate 1200

IP Configuration

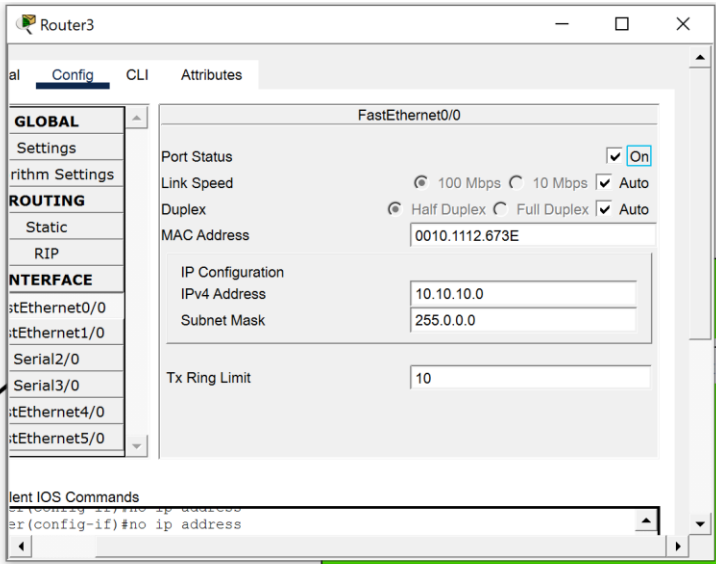
IPv4 Address 30.30.30.1

Subnet Mask 255.0.0.0

Tx Ring Limit 10

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Router3

Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

Serial2/0

Serial3/0

FastEthernet4/0

FastEthernet5/0

FastEthernet0/0

Port Status ☒ On

Link Speed ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☒ Half Duplex ☐ Full Duplex ☒ Auto

MAC Address 0010.1112.673E

IP Configuration

IPv4 Address 10.10.10.0

Subnet Mask 255.0.0.0

Tx Ring Limit 10

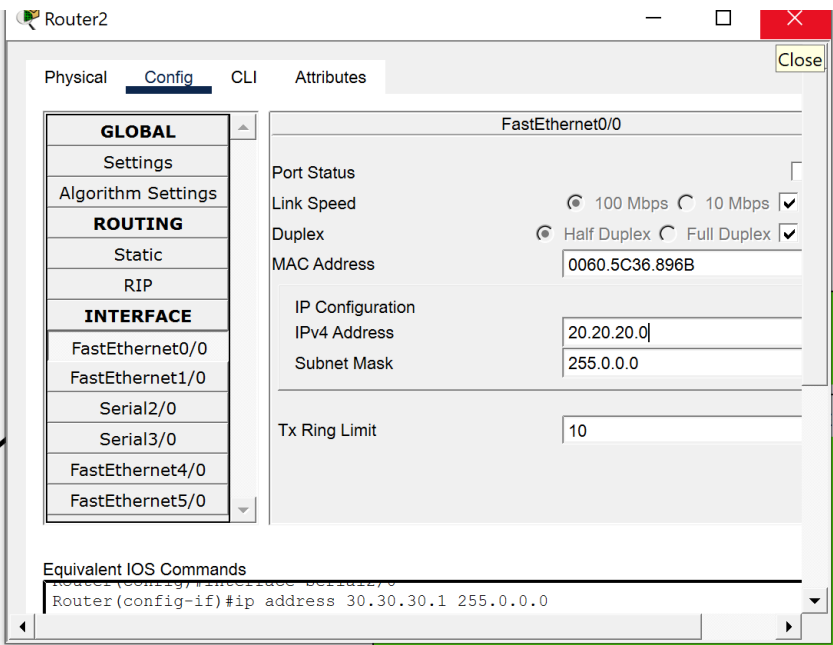
Equivalent IOS Commands

Router(config-if)#no ip address

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Router2

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

Serial2/0

Serial3/0

FastEthernet4/0

FastEthernet5/0

FastEthernet0/0

Port Status ☐

Link Speed ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☒ Half Duplex ☐ Full Duplex ☒ Auto

MAC Address 0060.5C36.896B

IP Configuration

IPv4 Address 20.20.20.0

Subnet Mask 255.0.0.0

Tx Ring Limit 10

Equivalent IOS Commands

Router(config-if)#ip address 30.30.30.1 255.0.0.0

PC-PT PC0

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PC-PT
PC0

Router3
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✕

al
Config
CLI
Attributes

GLOBAL

Settings

rithm Settings

ROUTING

Static

RIP

INTERFACE

tEthernet0/0

tEthernet1/0

Serial2/0

Serial3/0

tEthernet4/0

tEthernet5/0

Serial2/0

Port Status ☒ On
Duplex ☒ Full Duplex
Clock Rate 1200

IP Configuration
IPv4 Address 30.30.30.1
Subnet Mask 255.0.0.0

Tx Ring Limit 10

THE NETWORK
NETWORK



ent IOS Commands

```

er(config-if)#exit

```

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PT
:0

PC0
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□
✕

☐ DHCP
☒ Static

IPv4 Address 20.20.20.1
Subnet Mask 255.0.0.0
Default Gateway 20.20.20.254
DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic
☒ Static

IPv6 Address /

Link Local Address FE80::2E0:F7FF:FED2:98BC

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

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PC1

IP Configuration

Interface: FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 20.20.20.2

Subnet Mask: 255.0.0.0

Default Gateway: 20.20.20.0

DNS Server: 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::20D:BDFF:FEA2:8080

Default Gateway:

DNS Server:

802.1X

PC-PT PC0

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Router3

```
Router(config-if)#
Router(config-if)#ip nat inside source static 10.10.10.1 30.30.30.1
Router(config)#ip nat inside source static 10.10.10.2 30.30.30.1
Router(config)#
Router(config)#
Router(config)#interface fa0/0
Router(config-if)#ip nat inside
Router(config-if)#exit
Router(config)#
Router(config)#interface s2/0
Router(config-if)#ip nat outside
Router(config-if)#exit
Router(config)#
Router(config)#
Router(config)#ip route 0.0.0.0 0.0.0.0 30.30.30.2
Router(config)#
Router(config)#
Router(config)#do write
Building configuration...
[OK]
Router(config)#
Router(config)#
```

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PC-PT PC0

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 PC-PT
PC0

Router2

```
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#! Configure Serial interface (CORRECT)
Router(config)#interface serial2/0
Router(config-if)#ip address 30.30.30.2 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#! Add static route back to internal network
Router(config)#ip route 30.30.30.0 255.255.255.252 30.30.30.1
Router(config)#
Router(config)#! Save configuration
Router(config)#do write
Building configuration...
[OK]
Router(config)#
Router(config)#! Exit to see prompt
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

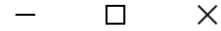
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**THE NETWORK
NETWORK**

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PC1



Physical Config **Desktop** Programming Attributes

Web Browser X

< > URL http://30.30.30.1 Go Stop

Cisco Packet Tracer

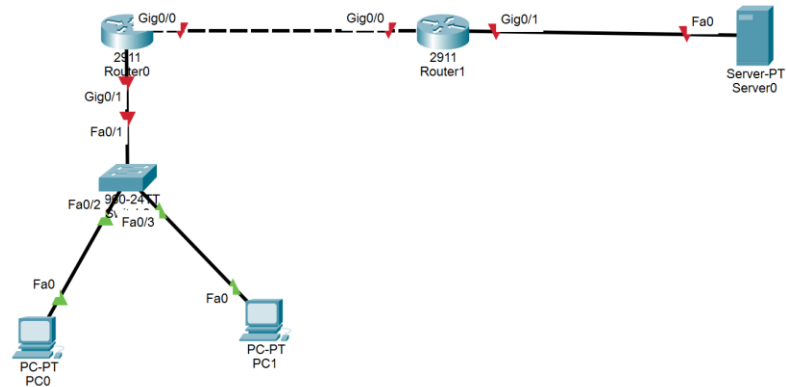
Welcome to Cisco Packet Tracer. Opening doors to new opportunities. Mind Wide Open.

Quick Links:
[A small page](#)

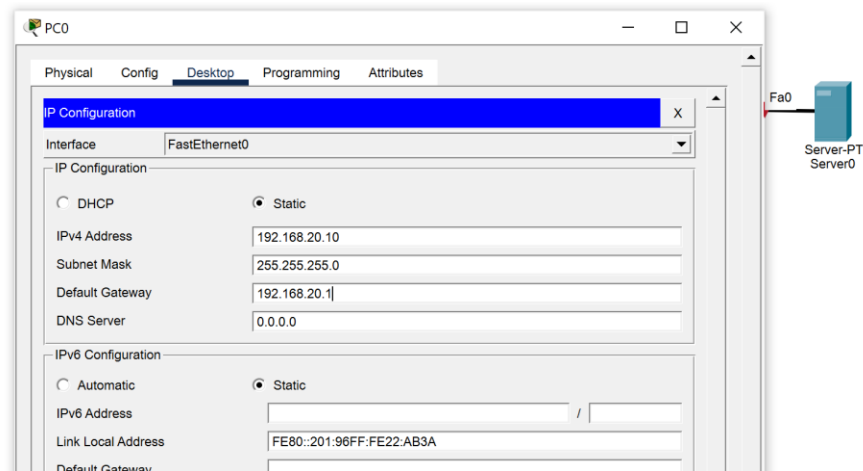
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Task 2: Dynamic NAT (10 marks)



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PC1

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.20.11

Subnet Mask 255.255.255.0

Default Gateway 192.168.20.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::201:C9FF:FECD:C516

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Root

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Server0

Physical Config Services **Desktop** Programming Attributes

IP Configuration X

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 8.8.8.8

Subnet Mask 255.0.0.0

Default Gateway 8.8.8.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::260:47FF:FE61:687

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Root

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Router0

Physical **Config** CLI Attributes

GLOBAL
 Settings
 Algorithm Settings
ROUTING
 Static
 RIP
SWITCHING
 VLAN Database
INTERFACE
 GigabitEthernet0/0
 GigabitEthernet0/1
 GigabitEthernet0/2

GigabitEthernet0/0
 Port Status ☒ On
 Link Speed ☐ 1000 Mbps ☒ 100 Mbps ☐ 10 Mbps ☒ Auto
 Duplex ☒ Half Duplex ☐ Full Duplex ☒ Auto
 MAC Address 00D0.580A.DE01
 IP Configuration
 IPv4 Address 200.100.50.1
 Subnet Mask 255.255.255.0
 Tx Ring Limit 10

Equivalent IOS Commands

Root 

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Router0

Physical **Config** CLI Attributes

GLOBAL
 Settings
 Algorithm Settings
ROUTING
 Static
 RIP
SWITCHING
 VLAN Database
INTERFACE
 GigabitEthernet0/0
 GigabitEthernet0/1
 GigabitEthernet0/2

GigabitEthernet0/1
 Port Status ☒ On
 Link Speed ☐ 1000 Mbps ☒ 100 Mbps ☐ 10 Mbps ☒ Auto
 Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto
 MAC Address 00D0.580A.DE02
 IP Configuration
 IPv4 Address 192.168.20.1
 Subnet Mask 255.255.255.0
 Tx Ring Limit 10

Equivalent IOS Commands

```

Router(config-if)#
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
  
```

Root 

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Router1

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

GigabitEthernet0/2

GigabitEthernet0/0

Port Status ☐ On

Link Speed ☐ 1000 Mbps ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 0002.166B.A801

IP Configuration

IPv4 Address 200.100.50.20

Subnet Mask 255.255.255.0

Tx Ring Limit 10

Equivalent IOS Commands

Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

School

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Share

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LAB 10

100% Windows (CRLF) UTF-8

Router1

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

GigabitEthernet0/2

GigabitEthernet0/1

Port Status ☒ On

Link Speed ☐ 1000 Mbps ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☒ Half Duplex ☐ Full Duplex ☒ Auto

MAC Address 0002.166B.A802

IP Configuration

IPv4 Address 8.8.8.1

Subnet Mask 255.0.0.0

Tx Ring Limit 10

Equivalent IOS Commands

Router(config)#interface g0/1
Router(config-if)#ip address 8.8.8.1 255.0.0.0
Router(config-if)#no shutdown

School

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Share

S

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I: Configured from console by console

Physical Config **CLI** Attributes

IOS Command Line Interface

```

Router(config)#interface GigabitEthernet0/1
Router(config-if)#ip address 192.168.20.1 255.255.255.0
Router(config-if)#ip address 192.168.20.1 255.255.255.0
Router(config-if)#
Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
enable
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 45
Router(config-router)#router-id 4.3.2.1
Router(config-router)#network 200.100.50.0 0.0.0.255 area 0
Router(config-router)#network 192.168.20.0 0.0.0.255 area 0
Router(config-router)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#00:13:39: %OSPF-5-ADJCHG: Process 45, Nbr 5.1.2.4 on
GigabitEthernet0/0 from LOADING to FULL, Loading Done
^
% Invalid input detected at '^' marker.
Router(config)#exitOSPF-5-ADJCHG: Process 45, Nbr 5.1.2.4 on GigabitEthernet0/0
from LOADING to FULL
^
% Invalid input detected at '^' marker.
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
          
```

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uring Dynamic NAT

ic NAT in your network, you will need to c
ie router which IPs from the internal netw
s out, simply run the following commands

Physical Config **CLI** Attributes

IOS Command Line Interface

```

Router(config-if)#no ip address
Router(config-if)#no ip address
Router(config-if)#ip address 200.100.50.20 255.255.255.0
Router(config-if)#ip address 200.100.50.20 255.255.255.0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state
to up
ip address 8.8.8.1 255.0.0.0
Router(config-if)#ip address 8.8.8.1 255.0.0.0
Router(config-if)#
Router(config-if)#enable
^
% Invalid input detected at '^' marker.
Router(config-if)#configure terminal
^
% Invalid input detected at '^' marker.
Router(config-if)#router ospf 45
Router(config-router)#router-id 5.1.2.4
Router(config-router)#network 200.100.50.0 0.0.0.255 area 0
Router(config-router)#network 8.8.8.0 0.0.0.255 area 0
Router(config-router)#exit
Router(config)#do write
Building configuration...
[OK]
Router(config)#
          
```

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PC0

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.
Request timed out.
Reply from 192.168.20.1: Destination host unreachable.

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

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Check Compatibility

Commands Group Centralized Deployment

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PC1

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Root

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Realtime

ARP Table for Router0

IP Address	Hardware Address	Interface
192.168.20.1	00D0.580A.DE02	GigabitEthernet0/1
192.168.20.10	0001.9622.AB3A	GigabitEthernet0/1
192.168.20.11	0001.C9CD.C516	GigabitEthernet0/1
200.100.50.1	00D0.580A.DE01	GigabitEthernet0/0
200.100.50.20	0002.166B.A801	GigabitEthernet0/0



